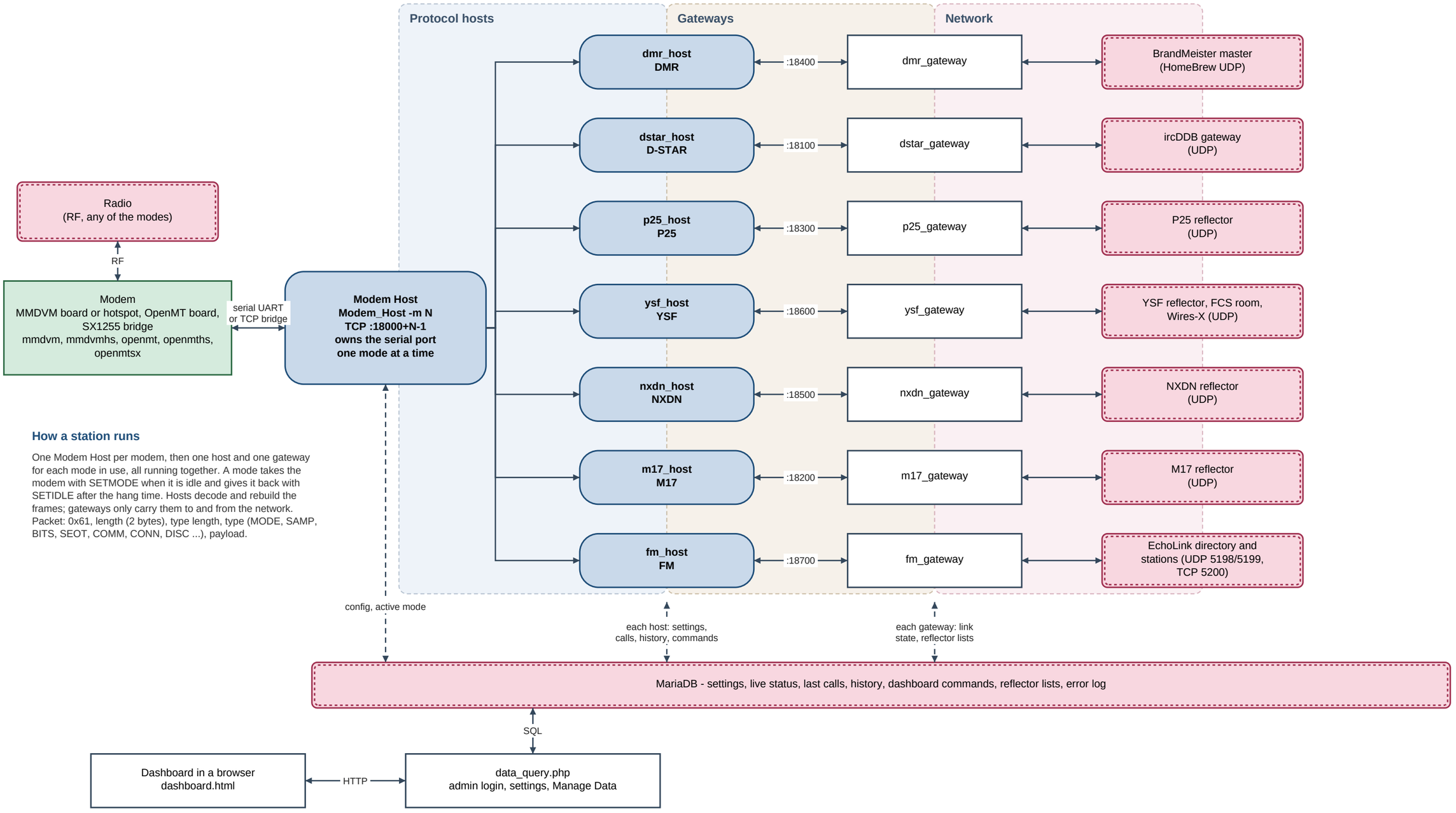
OpenMT - the whole system

Every program that runs on a station, and how data moves between them. N is the modem number (1 to 10); every port below is shown for modem 1.



How a station runs

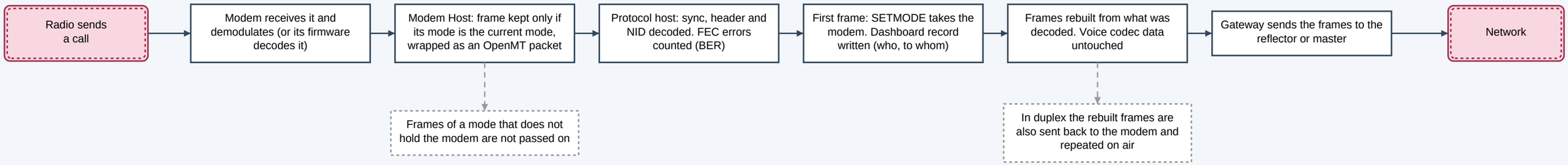
One Modem Host per modem, then one host and one gateway for each mode in use, all running together. A mode takes the modem with SETMODE when it is idle and gives it back with SETIDLE after the hang time. Hosts decode and rebuild the frames; gateways only carry them to and from the network. Packet: 0x61, length (2 bytes), type length, type (MODE, SAMP, BITS, SEOT, COMM, CONN, DISC ...), payload.

Reflector links are made from the dashboard: the command goes into the database, the host reads it, tells its gateway to link or unlink, and writes the result back.
With -m N each port is 18000+N-1 (host to Modem Host), 18100+N-1 (D-STAR), 18200+N-1 (M17), 18300+N-1 (P25), 18400+N-1 (DMR), 18500+N-1 (NXDN), 18600+N-1 (YSF), 18700+N-1 (FM).

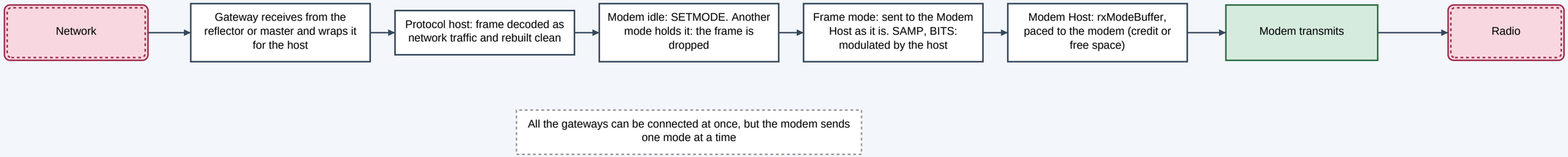
OpenMT - one call through the system

A call from the radio to the network, a call from the network to the radio, and how a call ends and gives the modem back.

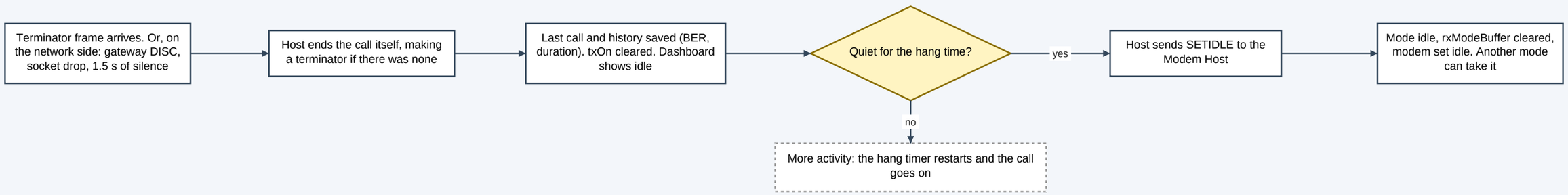
Radio -> network



Network -> radio



End of a call (either direction)



A transmission that runs too long is ended by the txTimeout in the Modem Host, which also sets the mode idle. Errors anywhere go to the error log table with reportError().